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Gender challenges in just energy transition: The labour market perspective

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ABSTRACT

The energy transition presents a significant challenge with broad societal and economic implications, offering an opportunity for a more sustainable, inclusive and equitable energy system. However, its success hinges on addressing social, economic, and cultural challenges beyond technology. Achieving a just transition requires equal opportunities for men and women in gender-biased male-dominated activities, impacting women's career paths. This study examines women's roles in energy transition activities in Spain, identifying participation gaps with a focus on horizontal segregation. It firstly designs a theoretical framework and operational delimitation for the energy transition sector. The working hypothesis suggests negative biases against women, affecting their entry into the labour market and contributing to horizontal segregation. Initial analysis reveals the severe under-representation of women in these activities across the EU and Spain. A logit model is used to analyse gender differences in the probability of accessing employment, controlling for personal and environmental characteristics. Additionally, a gap decomposition methodology examines the impact of various factors on gender differences. Results from the econometric model focused in Spain suggest that once controlled for personal characteristics, women have a lower probability of being employed in energy transition, varying by occupation. This could be interpreted as a working environment that discriminates women with entry barriers associated with worker sex, including those related to horizontal segregation. This work contributes with a novel comprehensive operational delimitation of the energy transition sector, expanding beyond renewable energies to incorporate key areas from national and international investment programs.

1. Introduction

The energy transition is a crucial challenge affecting all of society and the economy, rather than a specific sector of activity. It involves a structural shift from fossil-based systems of energy production, distribution and consumption to clean energy sources based on renewable energies and zero carbon emissions. In 2023, the energy sector led a record 18 % drop in emissions, compared to 2022, largely due to increased renewable electricity generation—especially from newly deployed solar and wind capacity [1]. Renewable sources reached 43 % of global electricity generation, with Chine (63 %), followed by the European Union (15 %) and the United States (7 %), accounting for 83 % of new solar PV capacity [2]. The European leadership was further reinforced in the first half of 2024, when wind and solar overtook fossil fuels in EU power generation, reflecting both the accelerating energy transition and growing concerns over fossil fuel supply security [3]. According to the latest data from International Renewable Energy Agency (IRENA) on total renewable energy capacity, Spain ranked as the second-leading country in the European Union in 2024, following Germany [4]. In 2024, over 55 % of the electricity generated in Spain came from renewable sources, with projections indicating an increase to 81 % by 2030, according to the National Integrated Energy and Climate Plan (NIECP). This progress positions Spain as a European leader in clean energy generation. Spain's influence extends beyond its national achievements, as its experience, abundant natural resources (sun and wind), and capacity for innovation make it a key actor in shaping the European energy agenda and guiding other countries through the energy transition.

Climate change and the ongoing energy transition are expected to lead significant social and labour market transformations. As economies

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shift away from fossil fuels towards low-carbon technologies, new employment opportunities will emerge in renewable energy, energy efficiency, and sustainable infrastructure. According to the IRENA, the renewable energy sector employed 13.7 million people, and it will create over 38 million jobs worldwide by 2030 and other related energy transition sector would account for 74.2 million jobs [5]. In the European Union, a recent study [6] showed that the installation of additional wind turbines and solar panels to deliver the EU Green Deal targets will require about 130,000 to 145,000 additional skilled workers in the EU by 2030. In Spain, the NIECP forecasts the creation of 250,000 to 350,000 new jobs annually, primarily high or medium-skilled.

A just transition must ensure equal job opportunities for all. In 2018, the Government of Spain and the International Labour Organization (ILO) signed a strategic partnership agreement to implement the Just Transition Guidelines. Spanish Just Transition Strategy builds upon this internationally endorsed framework to support the ecological transition while maximizing employment outcomes and promoting social and territorial justice and cohesion across the country [7]. It is important to note that low-emission energy systems are not inherently more inclusive than traditional systems [8]. Indeed, without corrective measures, gender-neutral interventions may end up benefiting groups that are already in a more privileged position, perpetuating structures of inequality in male-dominated energy systems [9]. Gender diversity is crucial, not only for social justice but also for economic reasons and the energy transition represents a significant opportunity for transformation towards a more sustainable, equitable, and inclusive energy paradigm, but its success depends on addressing these associated social and economic challenges.

The literature on women and energy is broad, but most of the research has focused on renewable energy sectors or decarbonized economic sectors from a gender perspective [8,10–12]. There is very little on the topic of just transition in energy sectors [13,14] and it is mainly focused on specific activities, mostly within the energy sector. Studies suggest that energy companies with greater gender diversity perform better [8]. The lack of women's inclusion could hinder the energy transition due to a shortage of qualified personnel [15]. But regardless that evidence, women have faced difficulties in being hired for energy transition-related activities within a context of gender sectoral segregation [16].

A key factor in that low participation has been women's historical lack of access to technical degrees, including STEM[1] university degrees and vocational training, which provide personnel for transition activities [17]. Despite women being a majority among graduates in the EU and in particular in Spain (in 2023 these figures were 58 % and 57 % respectively, for graduates aged 20–29), their participation in STEM related fields is still low. In the European Union (EU-27), in 2023 only 15.5 (per 1000) women in the 20–29 age group graduated in tertiary education in STEM fields, compared to 29.0 for men. The gap is wider in Spain, where those numbers are 13.1 and 33.5 (per 1000). Moreover, as a share of all tertiary education graduates in STEM education fields, the participation of women has remained stagnant around 33 % in the EU and 28 % in Spain, which means that only 1 out of every 3 graduates in STEM is a woman [18].

Within the labour market, the participation of women in energy transition employment is quite diverse, women hold 40 % of full-time positions in the solar photovoltaic (PV) sector, which is nearly double the percentage in the wind or oil and gas sectors, where they account for 21 % and 22 %, respectively [19]. Moreover, women are predominantly hired for administrative positions, where they make up 58 % of the workforce, while they are underrepresented in STEM positions (32 % of the workforce) and in non-STEM technical positions (35 % of the workforce). The situation is similar in Spain, where in 2010 women represent 26.6 % of the workforce in renewable energy [20] and 23.8 % in energy sector as a whole [21]. The segregation within energy sectors is also observed in Spain, where almost 64 % of women worked in administrative positions. Other areas such as "promotion/marketing/sales" and, to a lesser extent, "project development" and "research, development and innovation" show relatively low percentages (around 30 %). While the participation is even lower in areas directly related to industrial production and installation (15–16 %) [20].

Although the data are not favourable to female participation, the historical evolution seems to be in line with greater inclusion. There has been slight progress in the energy sector between 2008 and 2019 in the European Union, where the share of women increased from 20 % to 23 % [22], or from 23.8 % in 2010 to 29.4 % in 2020 in Spain [21].

In this context, the participation of women in the energy transition is a matter of national and international concern, both in political, economic, social, and academic spheres [23]. Incorporating a gender perspective in a just energy and environmental transition is essential to address existing and emerging inequalities and vulnerabilities, ensure and protect fundamental labour rights, and advance women's empowerment [24].

The aim of this paper is to understand the position and situation of women in employment related to the energy transition economic activity. For that purpose, we firstly use data for the European Union (EU) and specifically for Spain to quantify existing gaps and analyse their patterns. Secondly, we focus on Spain and apply logit models and decomposition techniques for those models to analyse the gender differences in the probability of accessing employment in the sector, controlling for different personal and environmental characteristics, including a detailed analysis of the factors influencing the observed differences.

In this study, energy transition economic activities are defined as those aligned with the transformation of the energy system towards net-zero carbon emissions, including renewable energy generation, energy efficiency, clean transportation, and related industrial and consulting services. This definition is operationalized using the National Classification of Economic Activities (NACE Rev. 2) [25], detailed in the Methodology section.

Our working hypothesis is that energy transition employment suffers from negative biases towards the employment of women, both in terms of their entry into the labour market and in relation to horizontal segregation. These biases are related to the social construction of the role of women in the private sphere, as presented in the theoretical framework.

This paper contributes to the literature in two ways. Firstly, by providing a comprehensive and operational delimitation of the energy transition sector in terms of economic activities and employment. One of the challenges in the analysis of gender gaps in this sector from an applied perspective is the lack of a delimitation of economic activities and employment related to energy transition, both from a conceptual and operational perspective. In this sense, we design an applied delimitation of the activities related to the energy transition using the NACE Rev. 2, that can be applied to any country that uses NACE, including the European Union. Therefore, this work contributes to the estimation of empirical evidence and thus, to the progress in comparative studies based on a broad and consistent definition. Secondly, the econometric results provide original evidence for Spain that allow the identification of recommendations to foster the gender sustainability of the employment opportunities in the sector. We focused on Spain, an interesting case in terms of gender equality. Spain has become one of the leaders in terms of commitment to gender equality in Europe [26] and it ranks fourth in the EU on the Gender Equality Index 2024.

[1] From a conceptual perspective STEM is associated with Science, Technology, Engineering and Mathematics, but operational definitions vary across studies and statistical sources. Eurostat uses UNESCO's International Standard Classification of Education (ISCED-F13) and includes in STEM: Natural sciences, mathematics and statistics; Information and Communication Technologies; and Engineering, manufacturing and construction.

2. Unwrapping gender differences in energy transition sectors? A theoretical framework

While significant progress has been made in reducing gender gaps in labour markets [27], there is growing concern about the stagnation of this convergence process. Gender disparities in specific economic sectors and occupations have become central to this concern, making labour market segregation a key aspect of those gaps [28]. Gender gaps in the labour market, and particularly those related to gender segregation, cannot be dissociated from the different dynamics that influence girls and women throughout their lives, affecting their educational attainment and labour market participation, as is the case in energy transition activities [29]. In the case of energy transition related sectors, women's participation can be understood within the broader context of persistent gender gaps in labour market participation, but it is also important to examine the specific barriers they encounter in male-dominated technically oriented fields [30], as energy transition is perceived as a technical field by society. In this regard, studies have suggested that women perceive academic careers related to energy sector as highly technical and physically demanding [31].

The explanation of the gender differences in educational attainment and labour market can be seen as the net result or combination of individual (internal) and social (external) spheres in the decisions of men and women, that overlap and interact in different ways [32]. In this process, individual decisions by girls and boys at educational stages are influenced by self-perception, which is associated with the social construction of gender roles and other social dynamics included in the so-called external factors [32]. Regarding science, the social construction of the field is not gender neutral, but is rather influenced by stereotypes [33]. At the core of the social construction that affects involvement of girls and women in STEM fields are the beliefs that boys are better than girls in science and maths and that STEM occupations are male fields [34]. This affects girls' self-perception negatively in comparison to boys in different aspects associated with higher interest and performance in those fields. As a result, the gap in enrolment rate (difference in the percentage of enrolment between men and women) in ICT fields and engineering have increased in the last years: 6.6 and 8.1 percentual points in 2014 and 2019 respectively in ICT, and 18.2 and 17.1 in 2014 and 2019 respectively in engineering. It is also true that not all the employment generated in energy transition requires STEM related education, but it is the common profile, and thus, "if women are to have a fair chance at occupying those roles in the future, they must also have greater access and incentive to enrol in engineering and STEM related subjects" ([35]: 23).

The structure of internal and external factors mentioned has a translation into the labour market explanations for gender differences in technical fields [36], including energy sectors [37]. This approach can be complemented by the global framework used to analyse gender gaps in labour market based on the determinants both from supply side (characteristics of the labour force) and the demand side (characteristics of the workplace and related actors).

Supply-side theories focus on the individual spheres and rely on the decisions that women make and the effects thereof on their labour conditions. Theories such as Human Capital Theory by Becker [38] study the impact of decisions in terms of education and human capital accumulation on labour conditions. Despite the fact that differences in education translate into disparities in the participation in the labour market, especially in terms of segregation in the field of study, subsequent literature stresses the limitations of these theories, acknowledging the key role society plays in shaping individual decisions [39]. In this sense, energy sectors are considered a male domain, highly demanding in terms of flexibility, mobility and work commitment, making it difficult to achieve work-family life balance, which overall has a discouraging effect on women's access and careers [40,41].

On the demand-side, theories have focused on discrimination in the labour market and on the structural factors that contribute to keep them in place, exploring how prejudices and stereotypes regarding women influence decisions at firms, especially when these decisions are conducted within male-dominated environments. Overall, these theories have a view of men as the desirable worker and women are considered less productive owing to household responsibilities [42,43], a perception that affects the allocation of responsibilities and power, formal and informal norms at work, selection and promotion processes and labour conditions [30,44].

Feminist, gender and sociological approaches go deeper into the roots of the structures that lies beneath both supply and demand type theories from a more global approach [45,46]. Within feminist theories, the identity theory of gender in the workplace expands the economic understanding of occupational segregation by explaining changes in segregation over time due to changes in societal notions of masculinity and femininity from a psychological and sociological perspective [44] and the undervaluation of women's jobs and occupations due to attitudes towards capacities and skills as well as employer and market practices [47]. Turnbull [48] suggests that jobs in the energy and related sectors were not originally designed for women, making them less suitable candidates by those responsible for hiring, who are mostly men. Moreover, there is a lack of information regarding employment opportunities, that might be offered through formal and organized forums or personal networks within the sector, where men have a comparative advantage over women, given that most of the workforce is male [10].

To summarize, conceptual models tend to support the idea that energy transition activities, being male-dominated and associated with a technological profile, pose extra challenges in the reduction of the existing gender gaps and require targeted policies to address the key determinants both at the educational and labour market stages. Thus, the conceptual framework supports our working hypothesis related to the negative biases towards the employment of women and their horizontal segregation within the energy transition activities.

3. Methodology and database

Despite the attention received from the policy perspective, there is no consensus about the delimitation of energy transition from the conceptual and applied perspective. Thus, one of the challenges of this work has been the design of that delimitation from an applied perspective. The objective, as with other concepts such as green economy, is to obtain an *ad hoc* delimitation of the activities related to the energy transition in terms of the economic sectors defined at the NACE Rev. 2, in order to use labour statistics to characterize employment.

The approach used in the delimitation relies on the activities that are at the core of the policies focused on energy transition. For the Spanish case, as mentioned previously the NIECP serves as the national strategic planning instrument that synthesizes energy and climate policies aiming to attain the decarbonization objectives established by the European Union. As such, it encompasses a comprehensive set of actions and investments, both public and private, to be developed along the upcoming decade. Consequently, it serves as a pertinent guide to address the primary inquiry: which subsectors or activities will concentrate employment generation associated with the energy transition. This is particularly noteworthy given the substantial investments anticipated in these subsectors, as indicated by the impact analysis within the NIECP itself.

Following an exhaustive examination of the array of actions outlined in the NIECP across its five axes of operation (Decarbonization, including renewable energies; Energy efficiency; Energy security; Internal energy market; and Research, innovation, and competitiveness), and building upon advancements in the realms of the green economy

and the so-called environmental sector, a tailored delineation of activities linked to the energy transition has been conducted, taking into account the limits of the labour force statistics in terms of the disaggregation provided for economic activities[2] [49]. These activities are categorized into five subsectors within the NACE Rev. 2 classification (Table 1), that distinguishes four core or characteristic subsectors (services primarily oriented towards emission reduction) and a subsector that encompasses activities termed as related (goods and services not specific to the energy transition but intricately connected to it, as they provide intermediate consumption or inputs to the core subsectors). Thus, the delimitation of energy transition goes beyond specific energy-related activities and includes those related to the changes in energy production, distribution, and consumption models to transform the current energy system, characterized by a high presence of fossil fuels, into a renewable energy-based system with net-zero carbon emissions.

Within these five subsectors, the associated occupations are diverse and span both STEM and non-STEM roles. Based on the International Standard Classification of Occupations, ISCO-08, major groups and available Labour Force Survey data, key occupation categories include: (i) Directors and managers (e.g., energy project managers, planning leads); (ii) Scientific and intellectual technicians and professionals (e.g., engineers, architects, economists, legal advisors in energy policy); (iii) Technicians and support professionals (e.g., lab technicians, IT specialists); (iv) Skilled workers in manufacturing and construction (e.g., electricians, renewable energy installers); (v) Office and administrative workers (e.g., financial and legal administrative roles). Many of these roles require backgrounds in STEM (science, technology, engineering, mathematics), but others rely on finance, law, urban and environmental planning, and similar disciplines. This mix underscores the multidimensional skill requirements of energy transition employment and the importance of addressing gender imbalances not only in technical fields, but also in the broader set of occupations critical to the transition.

Table 1
Delimitation of the energy transition in Spain's National Classification of Economic Activities (NACE Rev. 2).

NACE Rev. 2	Description
SUBSECTOR 1:	Electrical energy, power grids and self-consumption
201	Manufacture of basic chemicals, fertilisers and nitrogen compounds, plastics and synthetic rubber in primary forms
351	Electric power generation, transmission and distribution
422	Construction of utility projects
SUBSECTOR 2:	Energy renovation and the installation of heating and cooling equipment
432	Electrical, plumbing and other construction installation activities
433	Building completion and finishing
SUBSECTOR 3:	Energy efficiency in transport
291	Manufacture of motor vehicles
292	Manufacture of bodies (coachwork) for motor vehicles; manufacture of trailers and semi-trailers
293	Manufacture of parts and accessories for motor vehicles
491	Passenger rail transport
492	Interurban passenger rail transport
493	Other passenger land transport
SUBSECTOR 4:	Energy consulting
711	Architectural and engineering technical services and related technical consultancy
SUBSECTOR 5:	Related activities
162	Manufacture of products of wood, cork, straw and plaiting materials
251	Manufacture of structural metal products
261	Manufacture of electronic components and boards
271	Manufacture of electric motors, generators, transformers and electricity distribution and control apparatus
272	Manufacture of batteries and accumulators
279	Manufacture of other electrical equipment
281	Manufacture of general-purpose machinery
282	Manufacture of other general-purpose machinery
331	Repair of fabricated metal products, machinery and equipment

(Source: Own elaboration using NACE Rev. 2 classification.)

[2] The operative delimitation was designed jointly with the Institute for the Just Transition (Spanish Ministry for the Ecological Transition and the Demographic Challenge).

The source of data used for the analysis of the employment within energy transition activities in Spain is the Labour Force Survey for the European Union, with specific focus in Spain at the econometric section, with data for different years.

The quantitative strategy relies firstly on the analysis of the employment to identify existing patterns of employment and the gender gaps in those patterns, providing a framework for studying differences between men and women entering the labour market in energy transition activities. This analysis also enables the determination of the unconditional probability of entering the labour market. Subsequently, a logit regression model is employed to further examine variations in the probability of being employed in the energy transition sectors. The model allows the analysis of the influence of different factors, including sex as the key variable, to reflect possible different patterns.

$$P = \Pr\,[y = 1/X] = F(X'\beta) \tag{1}$$

where P is the probability of being employed in the analysed sectors, X is a vector of explanatory or control variables, and β is a vector of coefficients to be estimated. Moreover, $F(X'\beta)$ is the cumulative distribution function of the logistic function:

$$F(X'\beta) = [\exp(X'\beta)]/[1 + \exp(X'\beta)] \tag{2}$$

The dependent variable is a dichotomous variable, which takes value 1 if the worker is employed at energy transition sectors and 0 if he or she is employed in other sectors. The estimated coefficients using logit models cannot be directly interpreted as the magnitude of the change in the outcome probability that is associated with a one-unit change in the corresponding explanatory variable, and thus they only represent the direction of the relationship between the dependent variable and the explanatory variables. Therefore, the sign of the coefficients indicates the direction of the change in the likelihood or probability of entering the labour market. To allow for a better interpretation of the results, the marginal effect at the mean (MEM) of each estimated coefficient is calculated. In our model, the marginal effect indicates the change in the probability of being employed at the energy transition sectors as the corresponding explanatory variable increases in one unit, holding all other variables at their mean.

We consider three groups of explanatory variables: individual, job and establishment characteristics. The set of individual characteristic variables includes the sex of the worker, age as a proxy of work experience and its squared value, nationality, and the educational level divided into four categories (below primary education, primary and lower secondary education, upper secondary and professional education, university education). The sex of the worker is included by means of a dichotomous variable (female) which takes a value of one if the worker is female and zero otherwise. Among the set of job characteristics, we included information on occupation aggregated at the nine major groups according to the ISCO-08, the type of contract (permanent or fixed-term), and the type of working day (full-time or part time). The set of establishment characteristics includes the type of ownership (public or private) and the geographical location at the NUTS-3 level.

The sign of the coefficient for the female variables indicates whether female workers have a lower/greater probability of being employed at

the energy transition sectors. The corresponding marginal effect is the difference in the probability of being employed at the energy transition sectors between females and males.

Additionally, we apply the methodology proposed by Fairlie [50] for decomposing differences in logit models to conduct a detailed analysis of the factors influencing the differences between females and males in the probability of being employed in transition energy sectors. This approach draws upon the decomposition technique initially formulated by Blinder and Oaxaca [51,52] for linear models, which has found extensive application in labour market studies for studying wage differentials among distinct demographic groups. The methodological involves several steps. Initially, a baseline logit model for probability of being employed in transition energy sectors is estimated using pooled data from the two groups, females, and males. Afterwards, the fitted probability is calculated for each observation in the dataset using the estimated coefficients from the baseline model. Then, the mean probability of being employed in transition energy sectors is computed separately for females and males. Lastly, the difference in mean probabilities between the two groups is decomposed into components attributable to differences in observed characteristics (age, education, occupation…), which constitute the so-called explained component, and differences in the coefficients of the explanatory variables, which conform the so-called unexplained component. The unexplained component can be closely related to the presence of gender discrimination, but also to the possibility of omitted important variables when explaining the probability of being employed in energy transition sectors.

4. Results and discussion

4.1. Diagnosis of employment in the energy transition sector. Spain vs European Union

Between 2015 and 2021, more than two million net jobs have been created in the EU-27, representing a growth of 11.5 % in employment in sectors related to the energy transition, nearly three times higher than the overall employment growth rate (3.7 %) (Fig. 1). In Spain, the employment growth rate in transition sectors has been similar to the overall economy (10.9 %) and slightly lower than that observed in the EU-27 (10.6 %). Regarding female employment, job creation in energy transition sectors has grown at a faster rate than for men, both in Spain (28.3 % vs. 7.6 %) and in the EU-27 (21.7 % vs. 9.3 %), although these results are partly associated to a previously low presence of women in these subsectors.

Despite the observed increase in the new jobs occupied by women, their participation remains quite low, especially in comparison to the total economy (Fig. 2). Academic literature has analysed the thresholds that must be considered for assessing the participation level of women, including the potential effects of the limited presence of women in the workplace [53]. Two key levels of underrepresentation are identified to better understand the dynamics generated in work environments. The first one is the 15 % participation threshold, below which the environment can be particularly challenging for women. Below this level, excessive visibility, increased pressure stemming from multiple factors, the possibility of experiencing isolation, the need to conform to stereotypes, discrimination among women, and physical or psychological stress deteriorate the work environment for women and may even lead to their expulsion from their positions. If the presence of women manages to increase, another crucial milestone is the 30 % mark, identified in this context with the sociological concept of critical mass, surpassing which would favour an environment supporting a positive evolution of women's presence.

Following previous works, we consider the following categories in terms of female share in employment: Severely underrepresented (0 % to 15 %); Underrepresented without reaching a critical mass (16 % to 30 %); Underrepresented with a critical mass (31 % to 45 %); Balanced representation (46 % to 55 %); Moderately overrepresented (56 % to 70 %); Highly overrepresented (71 % to 85 %); and very highly overrepresented (86 % to 100 %). In 2021, women represented 16.5 % of total employment in Spain, below the EU27 average of 19.7 % for 2020 (Fig. 2). Considering that women's participation in employment in the overall economy was 46.2 % (46.1 % in the EU-27), the participation gap was 29.7 percentage points (26.4 for the EU-27). Thus, the participation of women in these sectors does not reach half of the average women's presence in overall employment. Despite the diversity of situations among EU-27 countries, in none of them women participation in energy transition surpasses the critical mass of 30 %. Women's participation in employment in the energy transition across the EU has increased in 2015-2020 by 1.6 percentage points, from 18.1 % to 19.7 %. Although the progress is positive, the pace of change is relatively slow and if that pattern persist, equality in access to jobs generated by

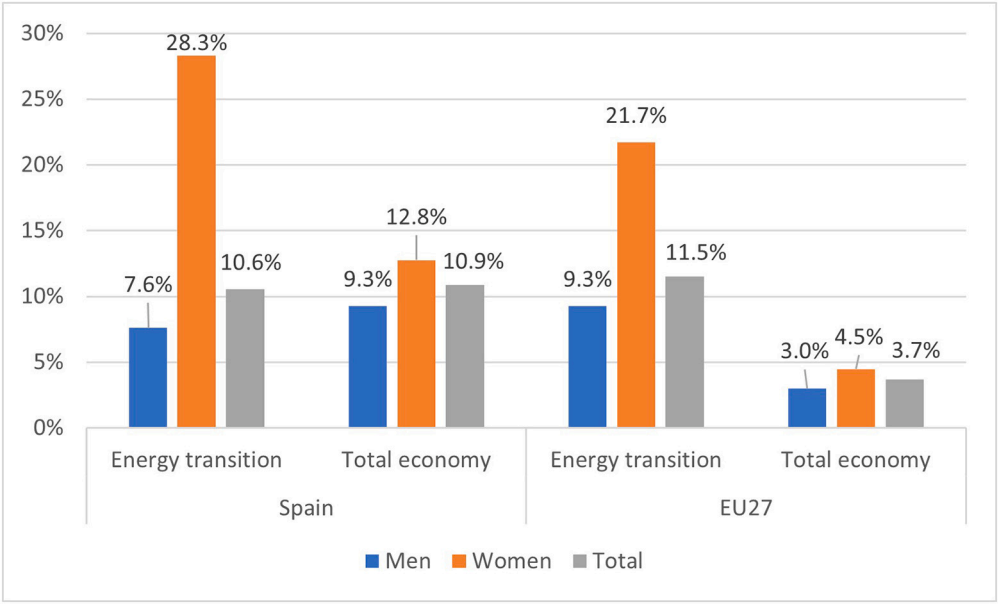

Fig. 1. Employment growth rate (2015–2021) by sex and activity.
(Source: Own elaboration from Martínez et al. ([49]:33).)

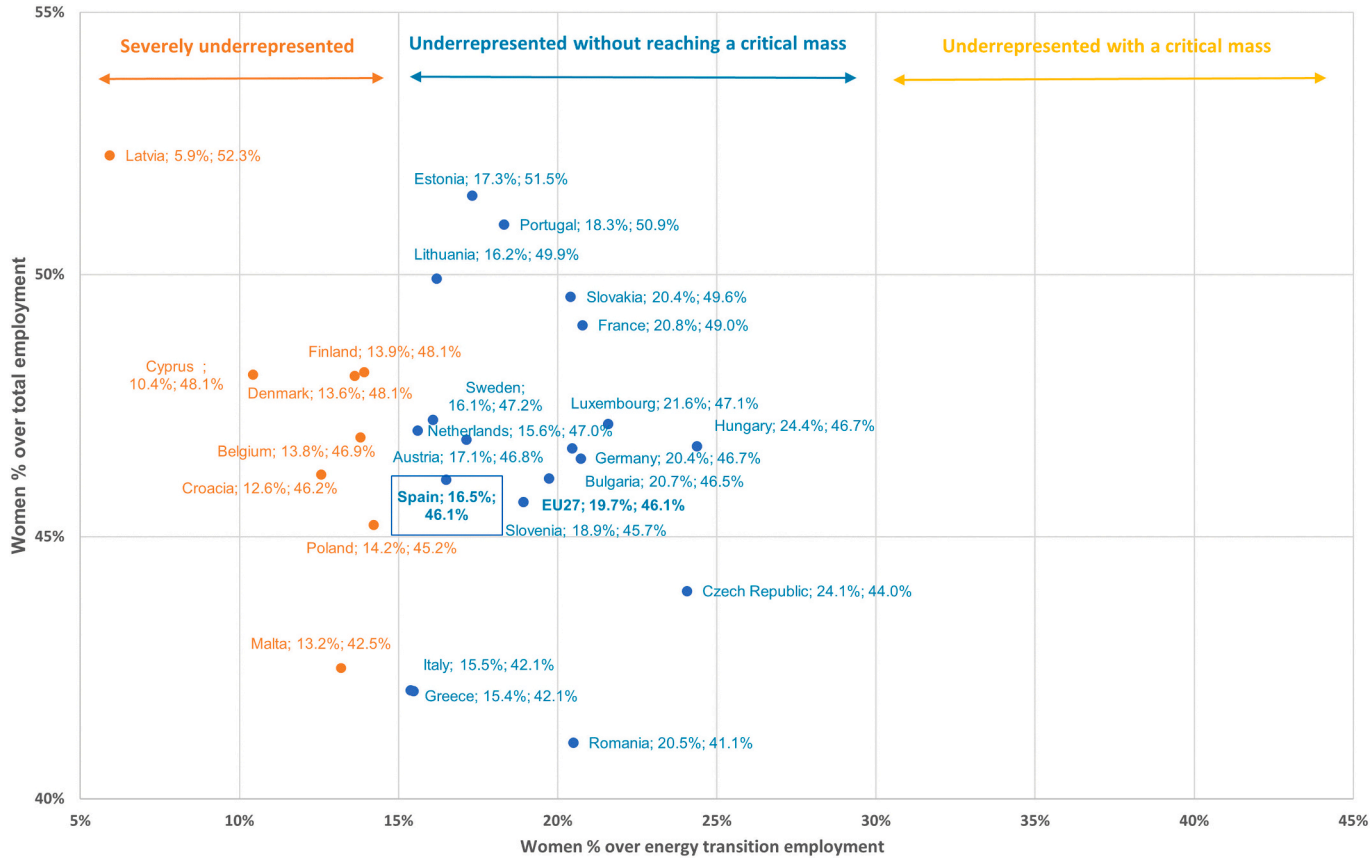

Fig. 2. Gender participation gap in employment in energy transition and in total economy (% of participation), 2021 (2020 for EU-27). (Source: Own elaboration from Martínez et al. ([49]:60-61))

the energy transition would be achieved well after the year 2050. By that time, most of the planned investments in different strategies, both at the national and European levels, targeting the energy transition will have already materialized.

4.2. Econometric results

The econometric analysis was carried using data from the Spanish Labour Force Survey, SLFS2022, (second quarter). The total number of observations in the sample is 52,680, where 4,550 correspond to the energy transition sectors and 48,130 to the rest of the economy. Taking into account the elevation factors provided in the SLFS, females represent 17.6 % of the sample in the energy transition sectors, and a 49.1 % in the rest of the economy (the distribution of the sample by different personal and firm characteristics is supplied as complementary material, Table A.1).

A summary of the estimation results of the logit model for the probability of being employed in the transition energy sectors are presented in Table 2. For clarity purposes, we only present the results for the female variable, age and its square, the level of education, and occupation. Column (1) shows the estimated coefficients and column (2) shows the estimated marginal effects at the mean (MEM). We also include the number of observations, the pseudo-R squared, and the Wald test for the null hypothesis of absence of joint statistical significance.

The estimated coefficient for the female variable is negative and statistically significant at the 1 % level, suggesting that female workers have a lower probability of being employed at the transition energy sectors than their male counterparts. More specifically, the marginal effect indicates that the probability of a female worker of being employed at the energy transition sectors is 2.6 percentage points lower than that of a male worker with similar characteristics. This result shows evidence that women face greater obstacles than men towards working in energy transition sectors, parts of which may be due to the presence of gender discrimination in these sectors.

Table 2
Estimation results for the logit model.

	Coefficient	MEM
	(1)	(2)
Female	−0.7839***	−0.0264***
Age	0.0351**	0.0012**
Age squared	−0.0005**	0.0000**
Primary and lower secondary education	0.8029*	0.0271*
Upper secondary and professional education	1.0122**	0.0341**
University education	0.8948**	0.0302**
Machinery operators	2.2261***	0.0750***
Skilled workers in manu. and const.	2.7014***	0.0910***
Skilled workers in the primary sector	−1.2760**	−0.0430**
Personal services, protection, sales workers	−0.7370***	−0.0248***
Accountants, office employees	1.4862***	0.0501***
Technicians and support professionals	1.5969***	0.0538***
Scientific and intellectual technicians	1.5513***	0.0523***
Directors and managers	1.3530***	0.0456***
N°	43,465	
R^2 (pseudo)	0.1961	
Wald test	2171.01	
(*p*-value)	(0.00)	

Notes: LFS grossing-up factors were used. Estimations also include a constant term, worker nationality, type of contract, type of working day and establishment regional location dummies (NUTS3). Statistical significance, with robust standard errors, is indicated by *** 1 %, ** 5 % and * 10 %. Marginal effects are calculated at the means (MEM).
(Source: Own calculations.)

Attending to the results for the other explanatory variables, both

experience, proxied by age, and the level of education increase the likelihood of being employed at the transition energy sectors. More specifically, the results indicates that an additional year of experience increases the probability by 0.1 percentage points. Also, workers with primary or lower secondary education have a probability of being employed in the transition energy sectors of 2.7 percentage points higher than those with an educational level below primary. For workers with upper secondary or professional education and university education, the probability is 3.4 and 3 percentage points higher respectively. Apart from "skilled workers from the primary sector" and "workers in personal services, protection, sales and catering", workers from any other occupation category are more likely to be employed in the energy transition sectors respect those in elementary occupations.

The results from the Fairlie decomposition for the gender differences in the probability of being employed in transition energy sectors are shown in Table 3. The mean probability of being employed in transition energy sectors is 12.4 % and 3.4 % for males and females respectively. Differences in observed characteristics between males and females explain almost two thirds (66.1 %) of the 9 percentage points of the observed gender difference. The variables which present the largest positive and statistically significant contribution are occupation and public firm. Differences in occupations between females and males explain half the difference (50.1 %) in the unconditional probability, indicating that women are less present in those occupations most in demand within the energy transition sectors. Data in the sample indicates that the main occupations in these sectors are "Skilled workers in manufacturing and construction industries" (37.6 % of total occupations), "Machinery operators" (19.6 %), "Scientific and intellectual technicians and professionals" (14.7 %) and "Technicians and support professionals" (12.1 %), while the presence of females is 1.7 %, 14 %, 23.3 % and 20.5 % respectively. Differences in the presence of females and males in public firms account for 11.5 % of the total difference in the unconditional probability, suggesting that women have a greater presence than men in public firms, which in turn have a very low presence in energy transition sectors. In most developed countries, the energy sector is dominated by private companies, although partial or full state ownership is often maintained in strategic areas such as electricity grids or gas pipelines. In Spain, since the late 20th century, the liberalization and privatization process has significantly transformed the national energy landscape. Formerly state-owned major energy producers have either become fully privatized or operate under mixed ownership structures [54].

Table 3
Decomposition results of differences in the probability of being employed in energy transition sectors.

	Estimated coefficient	Standard error	%
	(1)	(2)	(3)
Unconditional probability			
Male	0.1238		
Female	0.0318		
Difference	0.0919		
Explained	0.0608		66.1
Unexplained	0.0312		33.9
Explained component			
Age	−0.0037	(0.0052)	−4.1
Age squared	0.0032	(0.0052)	3.5
Education	0.0008	(0.0022)	0.9
Nationality	−0.0011***	(0.0004)	−1.2
Occupation	0.0461***	(0.0028)	50.1
Type of contract (permanent)	0.0003	(0.0003)	0.3
Type of working day (part time)	0.0024***	(0.0004)	2.6
Public	0.0106***	(0.0009)	11.5
Regional location NUTS3	0.0023***	(0.0003)	2.5

Notes: LFS grossing-up factors were used. Robust standard errors between parentheses. Statistical significance is indicated by *** 1 %, ** 5 % and * 10 %. (Source: Own calculations.)

The unexplained component represents a 33.9 % of the total difference in the unconditional probability, a component which accounts for gender discrimination and the omission of important factors (unobserved factors) when explaining the probability of being employed in energy transition sectors.

To gain further insights into the observed difference in the probability of being employed in the transition energy sectors between female and male workers, we replicated the estimations of the logit model for the major occupations groups (with the exception of skilled workers from the primary sector due to the low number of observations). A summary of the estimation results is presented in Table 4 (further results are provided as complementary material, Tables A.2 and A.3).

As it can be appreciated, the estimated coefficient for the female variable is negative and statistically significant at the 1 % level in five out of the eight occupation categories considered. The three occupations where the estimated coefficients are not statistically significant and thus, there are not significant differences in the probability of being employed in the energy transition sectors between females and males are "accountants and other office employees", "catering, personal services, protection and sales workers", and "plant and machinery operators and assemblers".

Among those occupations where female workers face obstacles to be employed in the energy transition sectors, the difference in probability is relatively high for "skilled workers in manufacturing and construction". The estimated marginal effect indicates that female skilled workers in manufacturing and construction have a probability of being employed in the energy transition sectors of 34.6 percentage points lower than a male worker in the same occupation group and similar characteristics. The difference in probability is also large in high skill occupations. For the category of "directors and managers" the estimated marginal effect suggests that the probability of being employed in the energy transition sectors is 8 percentage points lower for females than for males. This probability difference is 6.6 percentage points in the category of "scientific and intellectual technicians and professionals" where higher education degrees related to STEM fields are required, and 7 percentage points in the category of "technicians and support professionals". An explanation for these results could be the scarcity of women in higher education degrees related to STEM fields, which are essential competencies in specific job positions within the energy transition sector [10].

5. Discussion

The results presented in the previous section confirm the existence of significant gender disparities in employment related to energy transition sectors in Spain. Despite the increase in women's participation, the levels remain below 20 %, far from achieving gender parity. These sectors show a high degree of occupational segregation, with women concentrated in administrative and support roles and underrepresented in technical and high-skill occupations, such as scientific professionals or skilled construction workers. This is consistent with previous findings that characterize the energy sector as highly masculinized, with women's roles confined to peripheral areas such as administration, finance, or support services [37,40,55]. This occupational segregation persists despite technological advances, such as automation, that have reduced the physical demands traditionally associated with the sector [37].

The econometric analysis confirms that, after controlling for personal and job characteristics, women have a lower probability of being employed in energy transition sectors. The marginal effect of −2.6 percentage points indicates a persistent disadvantage for women, even when they have comparable profiles to men. This finding supports the working hypothesis regarding gender bias and horizontal segregation in these activities. The decomposition analysis further reveals that around two-thirds of the gender gap can be explained by observable characteristics, particularly occupation and employment in public firms. However, a significant share of the gap (approximately 34 %) remains

Table 4
Estimation results logit model by occupation. Coefficients and marginal effects for the female variable.

	Directors and managers	Scientific and intellectual technicians	Technicians and support professionals	Accountants, office employees	Personal services, protection, sales workers	Skilled workers in manu. and const.	Machinery operators	Elementary occupations
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
PANEL A: COEFFICIENTS								
Female	−1.0279***	−1.3248***	−0.8634***	0.2028	−0.1080	−1.8617***	−0.1700	−1.3079***
PANEL B: MEM								
Female	−0.0801***	−0.0662***	−0.0701***	0.0122	−0.0009	−0.3462***	−0.0239	−0.0220***
N°	908	8153	4940	5047	6366	4422	3835	4949
R^2 (pseudo)	0.1056	0.2212	0.1201	0.0573	0.1089	0.0639	0.0822	0.1472
Wald test	52.50	300.39	228.79	98.61	106.55	203.73	223.36	226.36
(p-value)	(0.06)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)

Notes: LFS grossing-up factors were used. Estimations also include a constant term, worker nationality, type of contract, type of working day and establishment regional location dummies (NUTS3). Robust standard errors between parentheses. Statistical significance is indicated by *** 1 %, ** 5 % and * 10 %. Marginal effects are calculated at the means (MEM).
(Source: Own calculations.)

unexplained. This residual may reflect discriminatory barriers, gendered expectations, or unmeasured factors such as informal networks or implicit biases in hiring and promotion.

These findings align with conceptual frameworks that highlight the influence of both internal and external factors in shaping gender disparities. On the one hand, women's underrepresentation in technical occupations reflects their historical exclusion from STEM education, where structural and cultural barriers persist [40,56]. The lower participation of women in science and technology fields is not due to aptitude but rather to gendered self-perceptions, limited mentoring, and a lack of female role models during secondary education [57]. These fields "provide students and graduates with the knowledge and skills required for subsequent employment in the energy sector or a position as a researcher in an energy-related field" ([58]: 19).

On the other hand, institutional contexts (including labour, tax, and family policies) continue to shape occupational choices and contribute to persistent segregation. These dynamics are part of what the literature conceptualizes as "gender regimes," which influence not only individual aspirations but also organizational norms and institutional structures [55].

Importantly, the analysis points to the heterogeneity of gender gaps across occupations. The largest differences are found in skilled manual jobs and high-level positions, such as managers and scientific professionals. These results reinforce the need for tailored policies that address not only access to employment but also career advancement. Prior research highlights that women remain significantly underrepresented in leadership and decision-making positions across the energy sector [40,55], which restricts their ability to shape the direction of the transition itself. This is especially concerning given evidence that the inclusion of women in decision-making improves the effectiveness and legitimacy of energy policies [59–61].

Moreover, the persistence of gender inequality in the energy transition reflects a broader gap between academic discourse and actual policy implementation. While there is a growing body of literature on energy justice and the gender-energy nexus, the integration of gender perspectives in concrete policy design remains limited [61,62]. This study contributes to bridging that gap by providing empirical evidence that can inform targeted interventions to promote gender equality in energy transition labour markets.

In sum, addressing the observed disparities requires comprehensive and coordinated strategies involving educational institutions, employers, and policymakers. Efforts should focus on dismantling gender stereotypes, promoting technical and scientific training for women, and ensuring gender-sensitive career development pathways. These actions are not only a matter of justice and inclusion but also essential to ensure that the energy transition is socially sustainable and aligned with broader sustainable development goals [59,60].

6. Conclusions

This study examined gender disparities in employment related to energy transition activities in Spain. Using a new operational delimitation of energy transition economic sectors based on NACE Rev. 2 and Labour Force Survey data, the research analysed the probability of employment in these sectors by sex using logit models and decomposition techniques. The findings support the initial hypothesis: women face systematic disadvantages in accessing employment in energy transition sectors, with significant horizontal segregation. These disadvantages persist even after controlling for individual and job characteristics.

Key conclusions are:

- Women's participation in energy transition jobs remains low, with levels below 20 %.
- The logit model shows that women are 2.6 percentage points less likely to be employed in the sector than men with similar profiles.
- Occupational segregation explains 50 % of the observed gender gap, while 34 % remains unexplained, possibly reflecting discrimination or unobserved barriers.
- Gender gaps vary across occupations, being most pronounced in technical and managerial roles.

The methodology, based on a novel operational definition of the sector and robust econometric techniques, enables a better understanding of the structural and compositional factors driving gender gaps. These results are in line with earlier literature that emphasizes the role of institutional contexts, gender regimes, and sectoral stereotypes in shaping occupational segregation and labour market inequalities [37,55,56].

This analysis offers evidence to inform targeted policies promoting gender equality in the labour dimension of the energy transition. It also addresses a key gap identified in the literature: the lack of empirical studies quantifying gender disparities in the energy transition at the national level [61]. By grounding its conclusions in robust data and highlighting structural barriers, this study contributes to the design of effective, inclusive policies that align with both energy justice and gender equality objectives [59,62].

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CRediT authorship contribution statement

Rosa Santero-Sánchez: Writing – review & editing, Writing –

original draft, Validation, Investigation, Conceptualization. **Belén Castro Núñez:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Víctor Martín Barroso:** Software, Methodology, Formal analysis. **Mª. Isabel Martínez Martín:** Conceptualization, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data were provided by Eurostat upon request and cant be shared.

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